

INFOSYS SPRINGBOARD

SMART URBAN MOBILITY & TRAFFIC INTELLIGENCE DASHBOARD

MILESTONE 4 – ADVANCED ANALYTICS

Microsoft Power BI Desktop

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Milestone 4 documentation prepared from the existing project model and mentor-specified dashboard structure.

Contents

- 1. Milestone 4 Overview and Objectives
- 2. Existing Data and Model Basis
- 3. Milestone 4 Analytical Measures and Derived Fields
- 4. Page 1 – Executive Intelligence Dashboard
- 5. Page 2 – Mobility Equity and Access Index
- 6. Page 3 – Demand Forecasting and Anomaly Detection
- 7. Interaction and Validation
- 8. Assumptions and Analytical Limitations
- 9. Conclusion
- 10. References

1. Milestone 4 Overview and Objectives

Milestone 4 extends the Smart Urban Mobility and Traffic Intelligence Dashboard into an advanced analytical layer while retaining the same Power BI project and underlying mobility model. The milestone focuses on mobility access, mobility equity, executive-level intelligence, demand forecasting and demand anomaly detection.

The work was developed using the existing mobility dataset rather than loading a new dataset. New Milestone 4 indicators are created from available fields using DAX calculated columns and measures.

1.1 Objectives

- Create a composite Mobility Access Index using accessibility, reliability and bike-availability components.
- Analyze mobility access across areas and demographic groups through comparative visuals and an equity matrix.
- Create an Executive Intelligence Dashboard with high-level mobility KPIs and supporting analytical visuals.
- Extend historical demand analysis into a Power BI forecasting view.
- Identify unusual demand observations using anomaly detection.
- Present the analysis through an interactive three-page Milestone 4 dashboard.

1.2 Milestone 4 Page Structure

Page	Title	Main Content
1	Executive Intelligence	Total Demand, Total Trips, Mobility Access Index, Average Reliability, Average Fare, demand trend, trips by transport mode, travel time by area, demand by location and reliability by mode.
2	Mobility Equity	Mobility Access Index, Average Reliability, Average Travel Time, Total Demand, Average Fare, access by area, demographic comparison, accessibility status and equity matrix.
3	Demand Forecast	Total Demand, historical demand forecast, demand category and demand anomaly detection.

2. Existing Data and Model Basis

Milestone 4 is built on the established Power BI semantic model from the earlier milestones. The central fact table is FactMobilityTable and it remains connected to Dim_Date, Dim_Location, Dim_Route, Dim_Vehicle and Dim_Weather using the existing one-to-many, single-direction relationships.

The documented project dataset contains 300 mobility records. The location dimension provides geographic attributes such as City, Zone, Location, Latitude, Longitude, Population, Population Density, Road Class and nearby-facility indicators. These existing fields provide the basis for the Milestone 4 access and equity analysis.

2.1 Existing Fields Used in Milestone 4

Source Area	Fields Used	Milestone 4 Role
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Dim_Date: Date — Time-series and forecasting axis

Dim_Location: Area, City, Zone, Location, Latitude, Longitude, Near_Hospital, Near_Market, Near_School — Access, equity and geographic analysis

Dim_Vehicle / FactMobilityTable: Vehicle_Type, Transport_Mode — Mode comparison

FactMobilityTable: Passenger_Count, Delay_Minutes, Travel_Time_Minutes, Average Fare, Bike Availability — Demand, reliability, travel-time and access indicators

FactMobilityTable: Demographic_Group, Accessibility_Status, Demand_Category — Milestone 4 segmentation and classification

3. Milestone 4 Analytical Measures and Derived Fields

The Milestone 4 calculations use the existing model and introduce advanced indicators needed by the mentor-specified dashboard. The important principle is that the Mobility Access Index combines three components on a common 0–100 scale: accessibility, reliability and bike availability.

3.1 Accessibility Score

Accessibility_Score is derived from the presence of nearby facilities already available in Dim_Location: a hospital, market and school. Each available facility contributes to the score, producing a 0–100 analytical accessibility indicator.

```
Accessibility_Score = ROUND(DIVIDE(Hospital + Market + School, 3) * 100, 2)
```

3.2 Accessibility Status

Accessibility_Status classifies the calculated accessibility score into an accessible or underserved category for the accessibility-status distribution visual.

```
Accessibility_Status = IF(Dim_Location[Accessibility_Score] >= 70, "Accessible", "Underserved")
```

3.3 Transport Mode

Transport_Mode is a derived analytical grouping created from the available vehicle categories so that the Executive Intelligence Dashboard can compare trips and reliability by transport mode.

```
Transport_Mode = SWITCH(TRUE(), VehicleType="Bus", "Bus", VehicleType="Motorcycle", "Cycling",  
VehicleType="EV", "Metro", VehicleType="Taxi", "Ride-Hailing", VehicleType="Car", "Ride-Hailing", "Other")
```

3.4 Demographic Group

Demographic_Group is an analytical grouping created for the mentor-specified equity comparison. It provides the categories used in the dashboard: Student, Senior, Family, Working Age and Low Income.

```
Demographic_Group = SWITCH(Transport_Mode, "Cycling", "Student", "Metro", "Working Age", "Ride-Hailing",  
"Family", "Bus", "Low Income", "Senior")
```

3.5 Equity Status

Equity_Status is derived from the accessibility score to support area and demographic equity analysis.

```
Equity_Status = IF(RELATED(Dim_Location[Accessibility_Score]) >= 70, "Equitable", "Needs Attention")
```

3.6 Demand Category

Demand_Category groups mobility observations into High, Medium and Low demand using the distribution of Passenger_Count in the current project dataset.

```
Demand_Category = SWITCH(TRUE(), Passenger_Count >= P75, "High", Passenger_Count <= P25, "Low", "Medium")
```

3.7 Average Reliability

For Milestone 4 executive analysis, Average Reliability is represented as a 0–100 score derived from the average-delay relationship. This keeps reliability comparable with the other normalized components of the access index.

```
Average Reliability = VAR AvgDelay = AVERAGE(FactMobilityTable[Delay_Minutes])  
VAR MaxDelay = MAXX(ALL(FactMobilityTable), FactMobilityTable[Delay_Minutes])  
RETURN IF(MaxDelay = 0, 100, (1 - DIVIDE(AvgDelay, MaxDelay, 0)) * 100)
```

3.8 Mobility Access Index

The Mobility Access Index is the central Milestone 4 composite measure. It combines the average accessibility score, reliability and bike availability after normalizing reliability and availability to a 0–100 scale.

```
Mobility Access Index = VAR Accessibility = AVERAGE(Dim_Location[Accessibility_Score])  
VAR Reliability = [Average Reliability]
```

```
VAR BikeAvailability = [Bike Availability % Measure] * 100  
RETURN DIVIDE(Accessibility + Reliability + BikeAvailability, 3)
```

A higher index represents a higher combined value across the three analytical components; a lower index indicates that one or more components contribute a lower score. The measure is intended for comparison under the active Power BI filter context.

4. Page 1 – Executive Intelligence Dashboard

The first Milestone 4 page is the Executive Intelligence Dashboard. It provides a high-level summary of mobility demand, trip activity, access, reliability and fare, followed by supporting visuals for trends, transport modes, area travel time, geographic demand and mode reliability.

4.1 KPI Cards

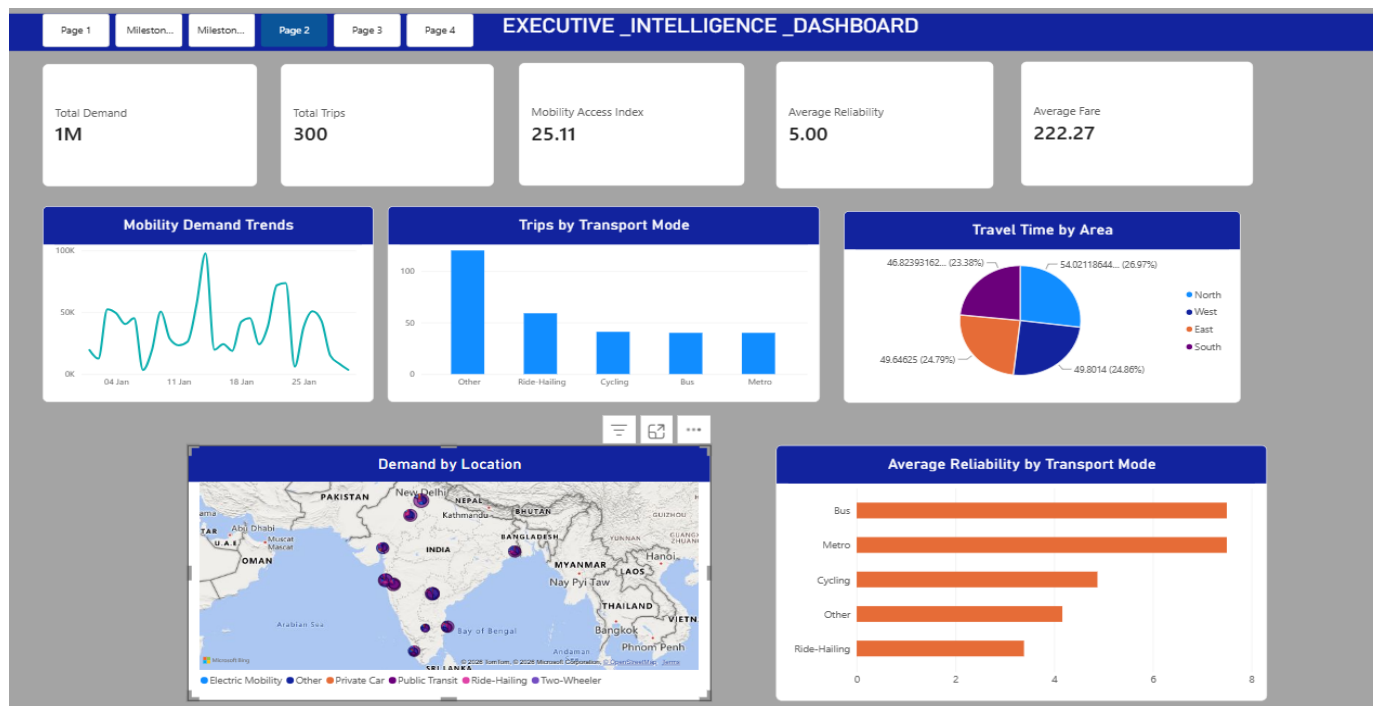
- Total Demand
- Total Trips
- Mobility Access Index
- Average Reliability
- Average Fare

4.2 Supporting Visuals

Visual	Type	Purpose
Mobility Demand Trends	Line chart	Show demand variation over date.
Trips by Transport Mode	Clustered column chart	Compare trip volume across transport modes.
Travel Time by Area	Donut/Pie chart	Compare travel-time distribution across areas.
Demand by Location	Map	Show geographic concentration of demand.
Average Reliability by Transport Mode	Bar chart	Compare reliability across transport modes.

4.3 Executive Interpretation

The page is designed as a top-to-bottom executive view. The KPI cards establish the current mobility state, the trend visual shows how demand changes over time, the mode chart shows trip composition, the area chart compares travel-time distribution, the map shows geographic demand, and the reliability chart compares service performance across transport modes.



5. Page 2 – Mobility Equity and Access Index

The second page focuses on mobility equity. It uses the Mobility Access Index as a common metric and compares access across areas and demographic groups while retaining the same project data and filter context.

5.1 KPI Cards

- Mobility Access Index
- Average Reliability
- Average Travel Time
- Total Demand
- Average Fare

5.2 Equity Visuals

Visual	Type	Purpose
Mobility Access by Area	Bar chart	Compare Mobility Access Index across areas.
Mobility Access Index by Demographic Group	Column chart	Compare access across the mentor-specified demographic groups.
Total Trips by Accessibility Status	Donut chart	Show the distribution of trips by accessibility status.
Mobility Access Equity Matrix	Matrix	Compare Mobility Access Index across area/zone and demographic group.

5.3 Filters

- Area
- Demographic Group
- Transport Mode

5.4 Equity Interpretation

The equity page supports comparison rather than a single aggregate view. Area-level access can be examined alongside demographic-group values, while the matrix provides a compact area-by-group view. The filters allow the analysis to be narrowed to a selected area, demographic group or transport mode.

Mobility Access Index
25.11

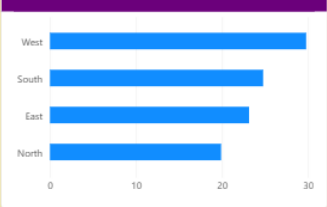
Average Reliability
5.00

Average Travel Time
49.46

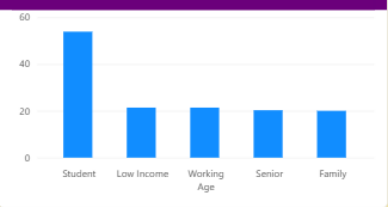
Total Demand
1M

Average Fare
222.27

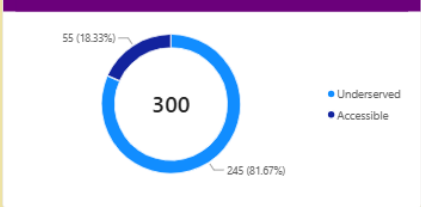
Mobility Access by Area



Mobility Access Index by Demographic Group



Total Trips by Accessibility Status



Mobility Access Equity Matrix

Zone	Family	Low Income	Senior	Student	Working Age	Total
East	14.81	23.15	14.81	48.15	25.93	23.15
North	16.67	14.82	16.48	48.15	14.82	19.90
South	18.52	18.52	20.48	53.70	18.52	24.79
West	24.91	29.52	24.29	59.57	28.22	29.79
Total	20.02	21.39	20.28	53.85	21.39	25.11

Area

- ☐ East
- ☐ North
- ☐ South
- ☐ West

Demographic Group

- ☐ Family
- ☐ Low Income
- ☐ Senior
- ☐ Student
- ☐ Working Age

Transport Mode

- ☐ Bus
- ☐ Cycling
- ☐ Metro
- ☐ Other
- ☐ Ride-Hailing

6. Page 3 – Demand Forecasting and Anomaly Detection

The third page extends the dashboard from descriptive demand reporting into forward-looking analysis. The page keeps Total Demand as the headline KPI, uses historical demand as the forecasting basis, classifies observations by demand category and highlights unusual demand behavior through anomaly detection.

6.1 Headline KPI

- Total Demand

6.2 Historical Demand Forecast

A line chart uses Dim_Date[Date] on the horizontal axis and Total Demand on the vertical axis. Power BI forecasting is applied to the historical series to produce a forward-looking forecast with a confidence interval.

Forecast setting used during the implementation: forecast length of 30 periods and a 95% confidence interval.

6.3 Demand Category

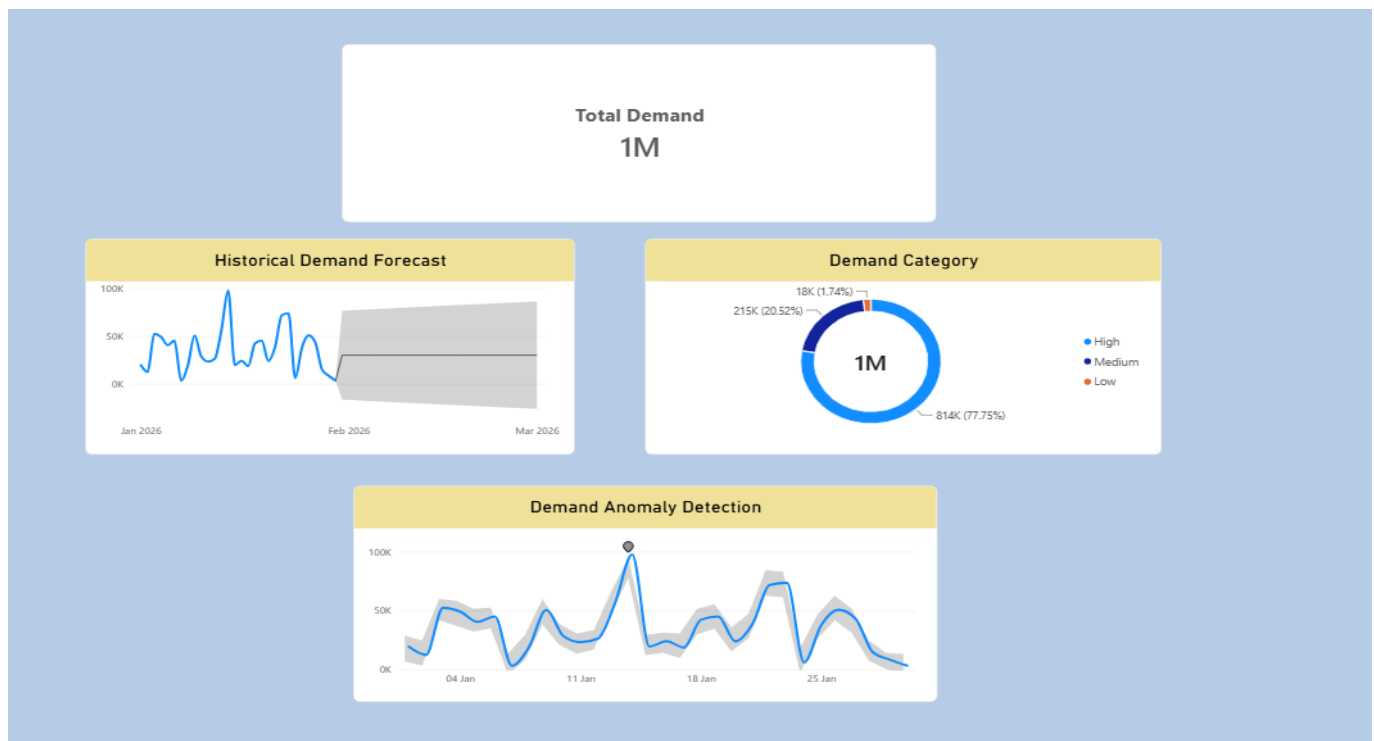
The Demand Category visual summarizes High, Medium and Low demand observations. This gives the executive page a simple classification layer alongside the continuous time-series view.

6.4 Demand Anomaly Detection

Anomaly detection is applied to the historical Total Demand line chart. Power BI identifies observations that depart unusually from the historical pattern, allowing spikes or drops to be highlighted without replacing the underlying demand trend.

6.5 Page 3 Analytical Workflow

- Read the Total Demand KPI.
- Inspect the historical demand trend.
- Use the forecast band to examine forward-looking demand behavior.
- Use Demand Category to understand the distribution of demand levels.
- Use anomaly markers to identify unusual demand observations.



7. Interaction and Validation

Milestone 4 continues the interactive behavior established in the earlier milestones. Measures recalculate in the current filter context, allowing the same dashboard to be examined by different areas, demographic groups, transport modes and dates.

7.1 Validation Checks

- Confirmed that the existing fact/dimension model remains the analytical foundation.
- Checked that KPI measures and derived indicators populate under the current filter context.
- Validated area and demographic-group comparisons in the equity page.
- Checked that the forecasting visual uses the date-based historical demand series.
- Applied anomaly detection to the historical demand series.
- Checked that the three Milestone 4 pages retain a consistent visual structure.

7.2 User Workflow

The intended workflow is: begin with executive KPIs, compare overall access and equity, then move to historical demand, forecast behavior and anomaly detection. This provides a progression from summary indicators to more detailed analytical evidence.

8. Assumptions and Analytical Limitations

Milestone 4 uses derived analytical fields where equivalent source fields are not directly available in the original mobility dataset. These fields are useful for dashboard analysis but should be interpreted as analytical constructs rather than direct survey or real-time operational measurements.

- Accessibility_Score is derived from nearby-facility indicators available in Dim_Location.
- Demographic_Group is an analytical grouping created for the mentor-specified equity analysis; it is not a direct demographic survey field in the source dataset.
- Accessibility_Status and Equity_Status are classification fields derived from the calculated access score.
- Mobility Access Index is a composite analytical index combining accessibility, reliability and bike availability.
- Bike availability remains the previously defined proxy based on the motorcycle-share measure; it is not real-time physical inventory.
- Demand forecasting is based on the historical demand series available in the project dataset.
- Demand anomaly detection identifies unusual observations relative to the historical series and should be interpreted within the available sample and time range.
- The Demand by Location map may remain environment-dependent in Power BI Desktop if the map visual/service is unavailable; the analytical intent and coordinate fields remain part of the design.

9. Conclusion

Milestone 4 extends the Smart Urban Mobility and Traffic Intelligence Dashboard into advanced analytics using the same Power BI model established in the earlier milestones. The milestone combines mobility access, equity analysis, executive intelligence, forecasting and anomaly detection within a structured three-page dashboard.

The Mobility Access Index provides a single composite indicator from accessibility, reliability and bike availability. The equity page enables area and demographic comparison, the executive page summarizes core mobility performance, and the forecasting page adds a forward-looking demand view together with anomaly detection.

Overall, Milestone 4 demonstrates how the existing mobility data model can be extended with DAX-based analytical indicators and Power BI advanced visuals to support a richer decision-support workflow.

10. References

- Microsoft Power BI Desktop documentation and visualization guidance.
- Microsoft DAX reference and calculation concepts.
- Microsoft Power Query documentation.
- Project Smart Urban Mobility dataset and validation workbook.
- Infosys Springboard Smart Urban Mobility project milestone documentation.
- Mentor-provided Milestone 4 dashboard structure and meeting walkthrough.